

Exercises 6

Exercise 1 (Galois Connection). Recall that a Galois connection is defined by two monotonic functions $\alpha : C \rightarrow A$ and $\gamma : A \rightarrow C$ between partial orders $\leq$ on C and $\sqsubseteq$ on A , such that

$$\forall a, c. \quad \alpha(c) \sqsubseteq a \iff c \leq \gamma(a) \quad (*)$$

1. Show that the condition $(*)$ is equivalent to the conjunction of these two conditions:

$$\forall c. \quad c \leq \gamma(\alpha(c)) \quad (1)$$

$$\forall a. \quad \alpha(\gamma(a)) \sqsubseteq a \quad (2)$$

2. Let $\alpha : C \rightarrow A$ and $\gamma : A \rightarrow C$ satisfy the conditions of a Galois connection. Show that the following three conditions are equivalent:
 - (a) $\alpha(\gamma(a)) = a$ for all a
 - (b) α is a surjective function
 - (c) γ is an injective function
3. State the condition for $c = \gamma(\alpha(c))$ to hold for all c . When C is the set of sets of concrete states and A is a domain of static analysis, is it more reasonable to expect that $c = \gamma(\alpha(c))$ or $\alpha(\gamma(a)) = a$ to be satisfied, and why?

Exercise 2 (lub and glb). Let $(A, \sqsubseteq)$ be a partial order such that every set $S \subseteq A$ has a greatest lower bound. Either prove that every set $S \subseteq A$ has a least upper bound, or show a counterexample.

Exercise 3 (Lattices). Consider the class of algebraic structures, called lattices, with signature $(\vee, \wedge)$, each of arity 2, satisfying the following axioms:

$$\begin{array}{ll|ll} x \vee y = y \vee x & (\vee\text{-Com}) & x \wedge y = y \wedge x & (\wedge\text{-Com}) \\ (x \vee y) \vee z = x \vee (y \vee z) & (\vee\text{-Assoc}) & (x \wedge y) \wedge z = x \wedge (y \wedge z) & (\wedge\text{-Assoc}) \\ x \vee x = x & (\vee\text{-Idem}) & x \wedge x = x & (\wedge\text{-Idem}) \\ x \vee (x \wedge y) = x & (\vee\text{-Abs}) & x \wedge (x \vee y) = x & (\wedge\text{-Abs}) \end{array}$$

1. Show that for any x and y , $x \wedge y = x$ if and only if $x \vee y = y$.
2. Define $x \leq y$ by $x \wedge y = x$. Show that the lattice equipped with $\leq$ is a partially ordered set.

3. Show that $\wedge$ and $\vee$ are respectively the binary *greatest lower bound* and *least upper bound* for $\leq$.

Exercise 4. Partitioning

1. Show that for any set S , $P_S = (2^S, \subseteq)$ is a lattice.
2. Consider a set $S = U \times V$. For each set $g \in 2^{U \times V}$, define $\bar{g} : U \rightarrow 2^V$ by $\bar{g}(u) = \{v \in V \mid (u, v) \in g\}$. Show that the bar function $\bar{\cdot}$ defines a bijection between $2^{U \times V}$ and $U \rightarrow 2^V$.
3. Consider the set of all functions $U \rightarrow 2^V$. Define a lattice on this set that is isomorphic to $(2^{U \times V}, \subseteq)$.

Exercise 5 (post). Let S be any set, $r \subseteq S \times S$ a binary relation, and $I \subseteq S$. Define $post : 2^S \rightarrow 2^S$ by $post(X) = I \cup r[X]$. Prove that $post$ is monotonic. Does $post$ admit a least fixed point?

Exercise 6. Let $f : [0, 1] \rightarrow [0, 1]$ such that

$$f(x) = \begin{cases} \frac{x^2}{4} + \frac{1}{10} & \text{if } x < \frac{1}{2} \\ \frac{1}{1+4^{-x}} & \text{if } x \geq \frac{1}{2} \end{cases}$$

Prove, using the fact that $([0, 1], \leq)$ is a complete lattice, that there exists $x \in [0, 1]$ such that $f(x) = x$.